SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	2024-IT	**Date:**	

Code of Conduct

I Sri Janani B(192421036) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Sri Janani B

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Given: Total spectrum = 45 MHz; bandwidth per simplex channel = 25 kHz.

Formula:

Simplex channels = Total bandwidth / Bandwidth per channel

Calculation:

Convert MHz into kHz:

45 MHz = 45 × 1000 = 45,000 kHz

Simplex channels = 45,000 ÷ 25

= 1800

Each full-duplex channel requires two simplex channels.

Full-duplex channels = 1800 ÷ 2

= 900

Answer: 900 full-duplex voice channels.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Given:

Total voice channels = 560

Number of cells = 7

Formula:

Channels per cell = Total channels ÷ Number of cells

Calculation:

Channels per cell = 560 ÷ 7

= 80

Answer: 80 channels per cell.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Given:

Number of carriers = 100

Time slots per carrier = 8

Formula:

Total voice channels = Number of carriers × Time slots per carrier

Calculation:

Total voice channels = 100 × 8

= 800

Answer: 800 simultaneous voice channels.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

Given:

Base station power = 80 W

Handset power = 4 W

Formula:

Power ratio (dB) = 10 × $\log_{10}$ (Base station power ÷ Handset power)

Calculation:

Power ratio = 10 × $\log_{10}$ (80 ÷ 4)

= 10 × $\log_{10}$ (20)

= 10 × 1.3010

= 13.01 dB

Answer: 13.01 dB.

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Given:

Carrier signal = −60 dBm

Interference signal = −92 dBm

Formula:

C/I (dB) = Carrier power − Interference power

Calculation:

C/I = −60 − (−92)

= −60 + 92

= 32 dB

Answer: 32 dB.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Given:

Total spectrum = 42 MHz

Bandwidth per channel = 300 kHz

Frequency reuse factor = 7

Formula:

Total channels = Total bandwidth ÷ Bandwidth per channel

Channels per cell = Total channels ÷ Reuse factor

Calculation:

Convert MHz into kHz:

42 MHz = 42 × 1000 = 42,000 kHz

Total channels = 42,000 ÷ 300

= 140

Channels per cell = 140 ÷ 7

= 20

Answer: 20 channels per cell.

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Given:

Initial received power = 64 W

Distance increase factor = 2

Path-loss exponent = 4

Formula:

New power = Initial power ÷ (Distance increase factor)^{n}

Calculation:

New power = 64 ÷ 2^{4}

= 64 ÷ 16

= 4 W

Answer: 4 W.

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Given:

Initial radius = R

New radius = 2R

Path-loss exponent = 2

Formula:

Power scaling factor = (New radius ÷ Initial radius)²

Calculation:

Power scaling factor = (2R ÷ R)²

= 2²

= 4

Answer: Transmit power must be increased by a factor of 4, or 4 times the original power.

Q9. Total Channels in a Cluster

Given:

Number of cells = 4

Channels per cell = 15

Formula:

Total cluster channels = Number of cells × Channels per cell

Calculation:

Total cluster channels = 4 × 15

= 60

Answer: 60 channels.

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Given:

Number of cells = 4

Channels per cell = 15

Formula:

Total cluster channels = Number of cells × Channels per cell

Calculation:

Total cluster channels = 4 × 15

= 60

Answer: 60 channels.

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Given:

Velocity = 100 km/h

Carrier frequency = 900 MHz

Speed of light = 3×10^8 m/s

Formula:

Doppler shift = (Velocity × Carrier frequency) ÷ Speed of light

Calculation:

Convert velocity into m/s:

100 × 1000 ÷ 3600 = 27.78 m/s

Doppler shift = $(27.78 \times 900 \times 10^6) \div (3 \times 10^8)$

= 83.34 Hz

Answer: Approximately 83.3 Hz maximum Doppler shift.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Given:

Total frame duration = 5 ms

Number of time slots = 8

Formula:

Time slot duration = Total frame duration ÷ Number of time slots

Calculation:

Time slot duration = 5 ÷ 8

= 0.625 ms

Answer: 0.625 ms per time slot.

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Given:

Total bandwidth = 24 MHz

Carrier spacing = 200 kHz

Formula:

Number of carriers = Total bandwidth ÷ Carrier spacing

Calculation:

Convert MHz into kHz:

24 MHz = 24 × 1000 = 24,000 kHz

Number of carriers = 24,000 ÷ 200

= 120

Answer: 120 carrier frequencies.

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Given:

Transmitted power = 8 W

Path loss = 33 dB

Formula:

Transmitted power (dBm) = 10 × $\log_{10}$ (Power in watts × 1000)

Received power = Transmitted power (dBm) − Path loss

Calculation:

Transmitted power = 10 × $\log_{10}$ (8 × 1000)

= 10 × $\log_{10}$ (8000)

= 39.03 dBm

Received power = 39.03 − 33

= 6.03 dBm

Answer: 6.03 dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Given:

Reference distance = 1 m

Distance = 2.5 km

Frequency = 1900 MHz

Formula:

FSPL (dB) = 32.44 + 20 $\log_{10}$ (Frequency in MHz) + 20 $\log_{10}$ (Distance in km)

Calculation:

Distance = 2.5 km

FSPL = 32.44 + 20 $\log_{10}$ (1900) + 20 $\log_{10}$ (2.5)

= 32.44 + 65.58 + 7.96

= 105.98 dB

Answer: Approximately 106 dB.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Given:

Number of cells = 6

Calls per cell = 250

Formula:

Total capacity = Number of cells × Calls per cell

Calculation:

Total capacity = 6 × 250

= 1500

Answer: 1500 simultaneous calls.

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Given:

Call arrival rate = 4000 calls/second

Average call duration = 3 minutes

Formula:

Active calls = Call arrival rate × Average call duration in seconds

Calculation:

Convert minutes into seconds:

3 × 60 = 180 seconds

Active calls = 4000 × 180

= 720,000

Answer: 720,000 active calls, assuming the arrival rate and average duration remain constant.

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Given:

Number of channels = 150

Bandwidth per channel = 25 kHz

Formula:

Total bandwidth = Number of channels × Bandwidth per channel

Calculation:

Total bandwidth = 150 × 25

= 3750 kHz

Convert kHz into MHz:

3750 ÷ 1000 = 3.75 MHz

Answer: 3.75 MHz.

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Given:

Number of adjacent frequency bands = 30

Guard band per gap = 40 kHz

Formula:

Total guard bandwidth = Number of guard bands × Guard bandwidth per gap

Calculation:

Assuming 30 guard-band gaps:

Total guard bandwidth = 30 × 40

= 1200 kHz

= 1.2 MHz

Answer: 1.2 MHz.

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Given:

Vehicle speed = 90 km/h

Cell radius = 1.5 km

Formula:

Cell diameter = 2 × Radius

Handoffs per hour ≈ Distance travelled per hour ÷ Cell diameter

Calculation:

Cell diameter = 2 × 1.5

= 3 km

Distance travelled in one hour = 90 km

Handoffs per hour ≈ 90 ÷ 3

= 30

Answer: Approximately 30 handoffs per hour.

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Given:

Regional centers = 8

Base stations per center = 40

Users per base station = 250

Formula:

Total users = Regional centers × Base stations per center × Users per base station

Calculation:

Total users = 8 × 40 × 250

= 320 × 250

= 80,000

Answer: 80,000 users.

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Given:

Total area = 400 km^2

Cell radius = 400 m = 0.4 km

Approximate cell area = R^2

Formula:

Number of cells = Total area ÷ Area of one cell

Calculation:

Area of one cell = 0.4^2

= 0.16 km^2

Number of cells = 400 ÷ 0.16

= 2500

Answer: Approximately 2500 cells.

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Given:

Total spectrum = 6 MHz

Bandwidth per channel = 30 kHz

Formula:

Number of channels = Total bandwidth ÷ Bandwidth per channel

Calculation:

6 MHz = 6000 kHz

Number of channels = 6000 ÷ 30

= 200

Answer: 200 channels.

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Given:

Number of subscribers = 1200

Calls per subscriber per hour = 4

Average call duration = 2.5 minutes

Formula:

Traffic (Erlangs) = Number of subscribers × Calls per hour × Call duration in minutes ÷ 60

Calculation:

Traffic = 1200 × 4 × 2.5 ÷ 60

= 12,000 ÷ 60

= 200 Erlangs

Answer: 200 Erlangs.

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Given:

Total bandwidth = 40 MHz

Bandwidth per user = 400 kHz

Formula:

Number of users = Total bandwidth ÷ Bandwidth per user

Calculation:

40 MHz = 40,000 kHz

Number of users = 40,000 ÷ 400

= 100

Answer: 100 simultaneous users.

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Given:

Number of base stations = 1200

Users per base station = 600

Formula:

Total user capacity = Number of base stations × Users per base station

Calculation:

Total user capacity = 1200 × 600

= 720,000

Answer: 720,000 simultaneous users.

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Given:

Total cells = 200

Cells paged per round = 4

Formula:

Paging rounds = Total cells ÷ Cells paged per round

Calculation:

Paging rounds = 200 ÷ 4

= 50

Answer: 50 paging rounds.

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Given:

Maximum simultaneous connections = 100,000

Call attempts = 40,000 per minute

Average call duration = 3 minutes

Formula:

Offered traffic = Call attempts per minute × Average duration in minutes

Blocked fraction = (Offered traffic − Capacity) ÷ Offered traffic

Calculation:

Offered traffic = 40,000 × 3

= 120,000 connections

Blocked connections = 120,000 − 100,000

= 20,000

Blocked fraction = 20,000 ÷ 120,000

= 0.1667

Blocked percentage = 0.1667 × 100

= 16.67%

Answer: Approximately 16.67% of the offered load is blocked under this simplified capacity model.

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Given:

Total subscribers = 600,000

Roaming percentage = 15%

Formula:

Roaming subscribers = Total subscribers × Roaming percentage ÷ 100

Calculation:

Roaming subscribers = 600,000 × 15 ÷ 100

= 90,000

Answer: 90,000 roaming subscribers.

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Given:

Maximum capacity = 12,000 calls/second

Current load = 9000 calls/second

Formula:

Utilization (%) = Current load ÷ Maximum capacity × 100

Calculation:

Utilization = 9000 ÷ 12,000 × 100

= 0.75 × 100

= 75%

Answer: 75% utilization.

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Given:

Total spectrum = 48 MHz

Bandwidth per channel = 200 kHz

Frequency reuse factor = 4

Formula:

Total channels = Total bandwidth ÷ Bandwidth per channel

Channels per cell = Total channels ÷ Reuse factor

Calculation:

48 MHz = 48,000 kHz

Total channels = 48,000 ÷ 200

= 240

Channels per cell = 240 ÷ 4

= 60

Answer: 60 channels per cell.

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Given:

Total bandwidth = 30 MHz

Channel spacing = 200 kHz

Formula:

Number of channels = Total bandwidth ÷ Channel spacing

Calculation:

30 MHz = 30,000 kHz

Number of channels = 30,000 ÷ 200

= 150

Answer: 150 channels.

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Given:

Total handoff attempts = 2500

Success rate = 99%

Formula:

Successful handoffs = Total attempts × Success rate ÷ 100

Calculation:

Successful handoffs = 2500 × 99 ÷ 100

= 2475

Answer: 2475 successful handoffs.

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Given:

Transmitted power = 12 W

Path loss = 45 dB

Formula:

Transmitted power (dBm) = 10 × $\log_{10}$ (Power in watts × 1000)

Received power = Transmitted power (dBm) − Path loss

Calculation:

Transmitted power = 10 × $\log_{10}$ (12 × 1000)

= 10 × $\log_{10}$ (12,000)

= 40.79 dBm

Received power = 40.79 − 45

= −4.21 dBm

Answer: −4.21 dBm.

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Given:

Number of cells = 7

Users per cell = 900

Formula:

Cluster capacity = Number of cells × Users per cell

Calculation:

Cluster capacity = 7 × 900

= 6300

Answer: 6300 simultaneous users.

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Given:

Frame duration = 4.8 ms

Number of time slots = 8

Formula:

Time slot duration = Frame duration ÷ Number of time slots

Calculation:

Time slot duration = 4.8 ÷ 8

= 0.6 ms

Answer: 0.6 ms per slot.

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Given:

Velocity = 150 km/h

Carrier frequency = 1.8 GHz

Speed of light = 3×10^8 m/s

Formula:

Doppler shift = (Velocity × Carrier frequency) ÷ Speed of light

Calculation:

Convert velocity into m/s:

150 × 1000 ÷ 3600 = 41.67 m/s

Doppler shift = $(41.67 \times 1.8 \times 10^9) \div (3 \times 10^8)$

= 250 Hz

Answer: Approximately 250 Hz maximum Doppler shift.

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Given:

Maximum simultaneous calls = 250,000

Call arrival rate = 70,000 per minute

Average call duration = 4 minutes

Formula:

Offered traffic = Call arrival rate × Average call duration

Blocked percentage = (Offered traffic − Capacity) ÷ Offered traffic × 100

Calculation:

Offered traffic = 70,000 × 4

= 280,000 calls

Blocked calls = 280,000 − 250,000

= 30,000

Blocked percentage = 30,000 ÷ 280,000 × 100

= 10.71%

Answer: Approximately 10.71% of the offered load is blocked under this simplified capacity model.

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Given:

Number of cells = 9

Users per cell = 600

Formula:

Cluster capacity = Number of cells × Users per cell

Calculation:

Cluster capacity = 9 × 600

= 5400

Answer: 5400 simultaneous users.

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Given:

Total calls processed = 60,000 calls/second

Dropped call percentage = 0.4%

Formula:

Dropped calls = Total calls × Dropped percentage ÷ 100

Calculation:

Dropped calls = 60,000 × 0.4 ÷ 100

= 240

Answer: 240 dropped calls per second.

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Given:

Transmitted power = 43 dBm

Received power = −87 dBm

Formula:

Path loss = Transmitted power − Received power

Calculation:

Path loss = 43 − (−87)

= 43 + 87

= 130 dB

Answer: 130 dB.

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Given:

Area covered by one base station = 2.5 km²

Number of base stations = 150

Formula:

Total coverage area = Area per base station × Number of base stations

Calculation:

Total coverage area = 2.5 × 150

= 375 km²

Answer: 375 km².

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Given:

Total spectrum = 48 MHz

Bandwidth per channel = 200 kHz

Frequency reuse factor = 6

Formula:

Total channels = Total bandwidth ÷ Bandwidth per channel

Channels per cell = Total channels ÷ Reuse factor

Calculation:

48 MHz = 48,000 kHz

Total channels = 48,000 ÷ 200

= 240

Channels per cell = 240 ÷ 6

= 40

Answer: 40 channels per cell.

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Given:

Total subscribers = 150,000

Users processed per cycle = 6000

Formula:

Paging cycles = Total subscribers ÷ Users per cycle

Calculation:

Paging cycles = 150,000 ÷ 6000

= 25

Answer: 25 paging cycles.

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Given:

Call duration = 9 minutes

Handoff interval = 3 minutes

Formula:

Number of handoffs ≈ Call duration ÷ Handoff interval

Calculation:

Number of handoffs ≈ 9 ÷ 3

= 3

Answer: Approximately 3 handoffs, assuming a handoff is counted at each 3-minute interval.

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Given:

Signal power = −85 dBm

Noise power = −115 dBm

Formula:

SNR (dB) = Signal power − Noise power

Calculation:

SNR = −85 − (−115)

= −85 + 115

= 30 dB

Answer: 30 dB.

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Given:

Number of users = 250

Required data rate per user = 150 kbps

Formula:

Total required data rate = Number of users × Data rate per user

Calculation:

Total required data rate = 250 × 150

= 37,500 kbps

Convert kbps into Mbps:

37,500 ÷ 1000 = 37.5 Mbps

Answer: Minimum aggregate data rate = 37.5 Mbps. The actual radio bandwidth required depends on spectral efficiency.

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Given:

Number of subscribers = 1500

Calls per subscriber per hour = 4

Average call duration = 2.5 minutes

Formula:

Traffic (Erlangs) = Subscribers × Calls per hour × Call duration in minutes ÷ 60

Calculation:

Traffic = 1500 × 4 × 2.5 ÷ 60

= 15,000 ÷ 60

= 250 Erlangs

Answer: 250 Erlangs.

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Given:

Number of control channels = 250

Bandwidth per channel = 12 kHz

Formula:

Total bandwidth = Number of channels × Bandwidth per channel

Calculation:

Total bandwidth = 250 × 12

= 3000 kHz

Convert kHz into MHz:

3000 ÷ 1000 = 3 MHz

Answer: 3 MHz.

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Given:

Number of simultaneous connections = 1500

Average data rate per connection = 600 kbps

Formula:

Total throughput = Number of connections × Data rate per connection

Calculation:

Total throughput = 1500 × 600

= 900,000 kbps

Convert kbps into Mbps:

900,000 ÷ 1000 = 900 Mbps

Answer: 900 Mbps.

SQ50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Given:

Tower coverage radius = 4 km

Total service area = 350 km²

$\pi = 3.1416$

Formula:

Area of one tower's coverage = $\pi \times R^2$

Number of towers = Total service area ÷ Area of one tower's coverage

Calculation:

Area of one tower's coverage = 3.1416×4^2

$= 3.1416 \times 16$

$= 50.2656$ km²

Number of towers = $350 \div 50.2656$

$= 6.96$

Rounding up to the next whole tower:

= 7 towers

Answer: Approximately 7 towers, assuming ideal circular coverage without overlap or coverage gaps.